สมาชิก

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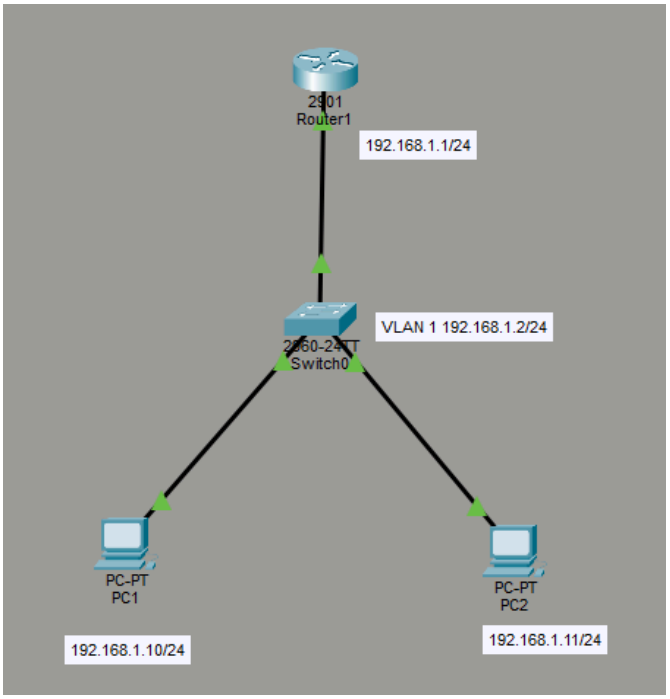
Lab 1: Complete Lab-Focused Teaching Package

🎯 LAB COMPLETION CRITERIA

- ☑ All devices respond to ping from at least one other device
- ☑ ARP tables show correct IP-MAC mappings
- ☑ Switch MAC table shows both PCs' MAC addresses
- ☑ Simulation clearly shows encapsulation process
- ☑ All troubleshooting scenarios resolved
- ☑ Worksheets completed with accurate data

PART 1: DEVICE CONFIGURATION (R1,S1,PC1,PC2)



📋 ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default Gateway	MAC (Dynamic)
R1	G0/0	192.168.1.1	255.255.255.0	N/A	0001.9643.AC01
S1	VLAN 1	192.168.1.2	255.255.255.0	192.168.1.1	0002.1745.8B02
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1	0050.7966.6803
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1	0090.2B14.7C04

PART 2: CONNECTIVITY VERIFICATION

PC1 ping PC2

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=4ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

PC1 ping R1

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=2ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

PC1 ping S1

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time=3ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

PC1

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           000c.cf35.9101        dynamic
  192.168.1.2           0001.96ce.5bcd        dynamic
  192.168.1.11          000a.41e1.c0d7        dynamic
```

PC2 ping PC1

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=5ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms
```

PC2 ping R1

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC2 ping S1

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC2

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           000c.cf35.9101        dynamic
  192.168.1.2           0001.96ce.5bcd        dynamic
  192.168.1.10          0060.2f35.e859        dynamic
```

R1

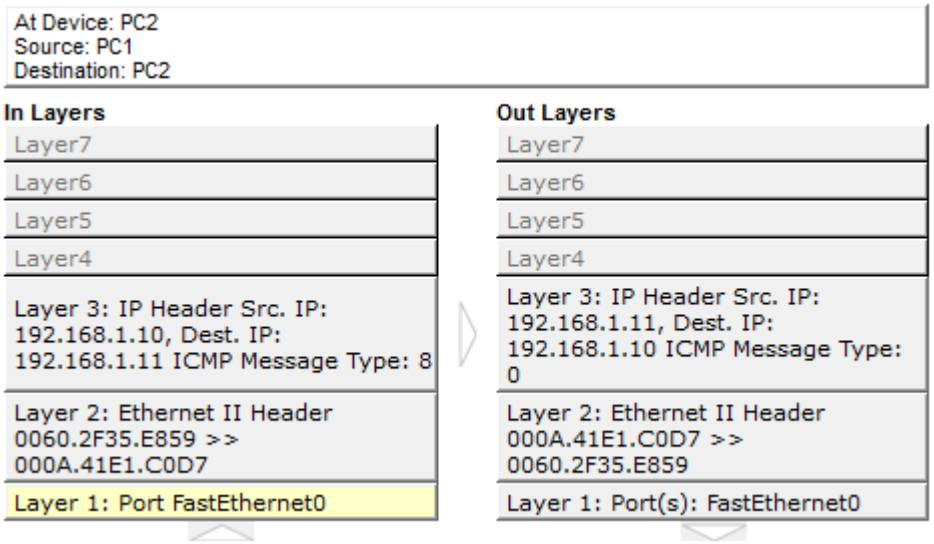
```
R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1>

R1>show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.1            -    000C.CF35.9101  ARPA   GigabitEthernet0/0
Internet  192.168.1.10          13    0060.2F35.E859  ARPA   GigabitEthernet0/0
Internet  192.168.1.11           3    000A.41E1.C0D7  ARPA   GigabitEthernet0/0
```

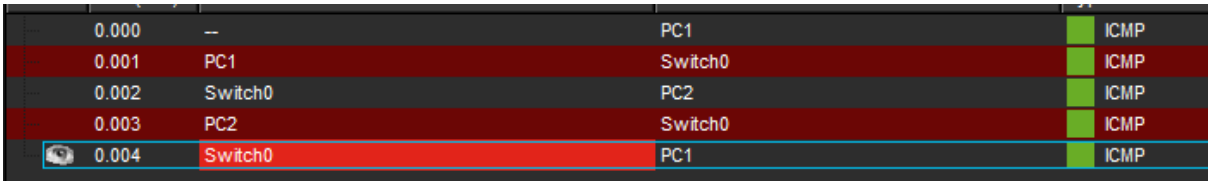
S1

```
s1>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- ---------
------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                               Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                               Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                               Fa0/13, Fa0/14, Fa0/15,
Fa0/16
                                               Fa0/17, Fa0/18, Fa0/19,
Fa0/20
                                               Fa0/21, Fa0/22, Fa0/23,
Fa0/24
                                               Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
s1>
```

```
s1>show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000c.cf35.9101    DYNAMIC     Gig0/1
s1>
```

PART 3: ENCAPSULATION ANALYSIS





คำอธิบาย : PC1 ส่ง ICMP Echo Request ไปยัง PC2 มี IP ต้นทาง 192.168.1.10 และ IP ปลายทาง 192.168.1.11

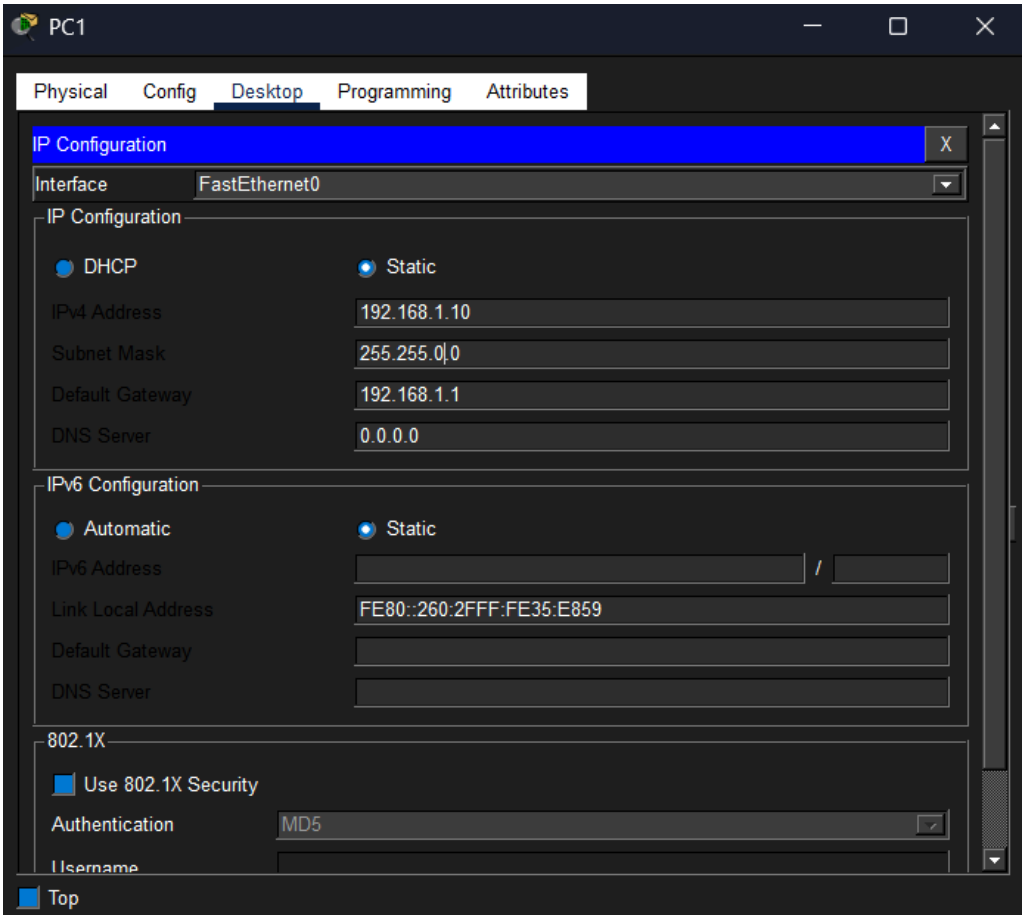
ก่อนส่ง ICMP จะมีการใช้ ARP เพื่อค้นหา MAC Address ของ PC2

เมื่อทราบ MAC แล้ว จึงห่อหุ้มข้อมูลใน Ethernet Frame และส่งผ่านสวิตช์ S1 ไปยัง PC2

Step	Protocol	OSI Layer	TCP/IP Layer	Key Field Observed
1	ARP Request	Layer 2	Network Access	Target IP: 192.168.1.11
2	ARP Reply	Layer 2	Network Access	Sender MAC: [PC2's MAC]
3	ICMP Echo	Layer 3	Internet	Type: 8, Code: 0
4	Ethernet	Layer 2	Network Access	Frame Check Sequence
5	ICMP Reply	Layer 3	Internet	Type: 0, Code: 0

PART 4: TROUBLESHOOTING CHALLENGE

Scenario A: Incorrect Subnet Mask




```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::260:2FFF:FE35:E859
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.10
   Subnet Mask.....................: 255.255.0.0
   Default Gateway.................: ::
                                     192.168.1.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

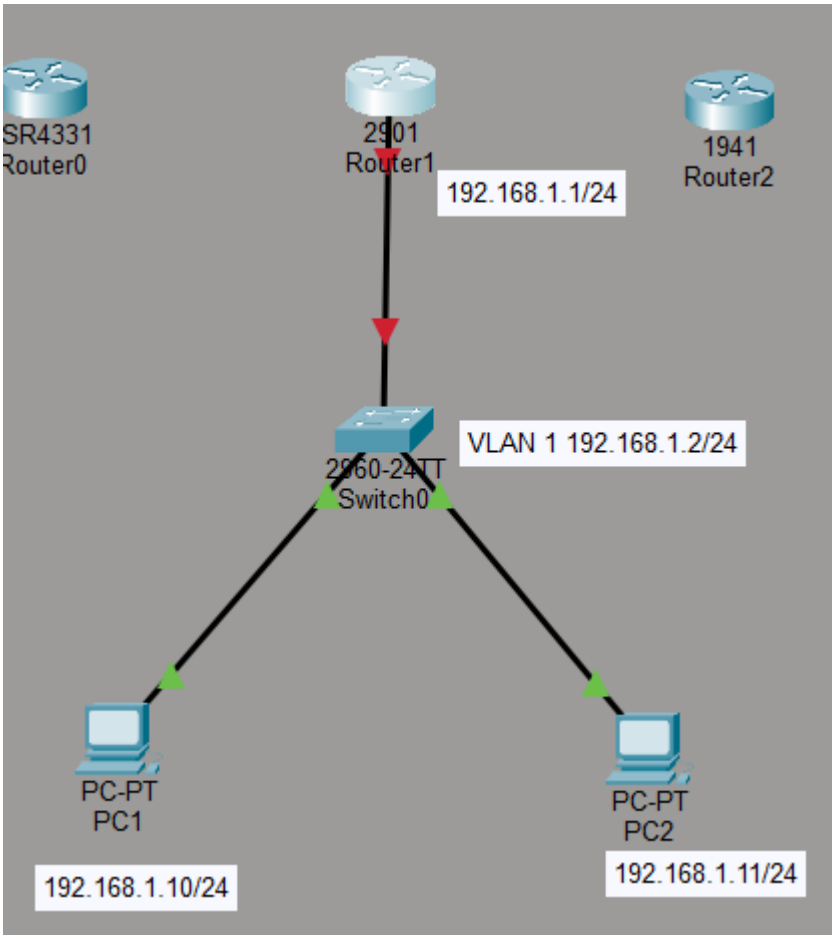
```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.11          000a.41e1.c0d7        dynamic
```

อธิบาย : ถึง subnet mask จะไม่ถูกต้องแต่ยังแก้ไข arp ได้เพราะอยู่ในเครือข่ายเดียวกัน

Scenario B: Router Interface Shutdown

ปิดพอร์ตที่ router




```
R1>enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface g0/0
R1(config-if)#shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to
administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to down

R1(config-if)#
```

Ping PC1 to R1

```
C:\>
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

arp check

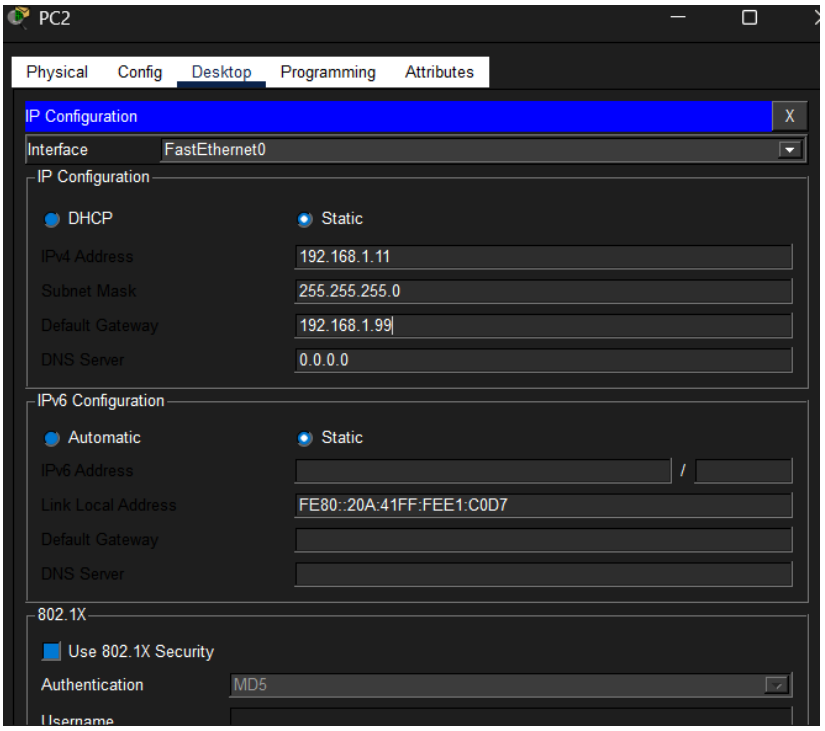
```
C:\>arp -a
No ARP Entries Found
C:\>
```

เปิดพอร์ต router

```
R1(config-if)#no shutdown

R1(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down

R1(config-if)#
```

Scenario C: Wrong Default Gateway

เปลี่ยน default gateway



```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::20A:41FF:FEE1:C0D7
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.99

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

ping PC2 to R1

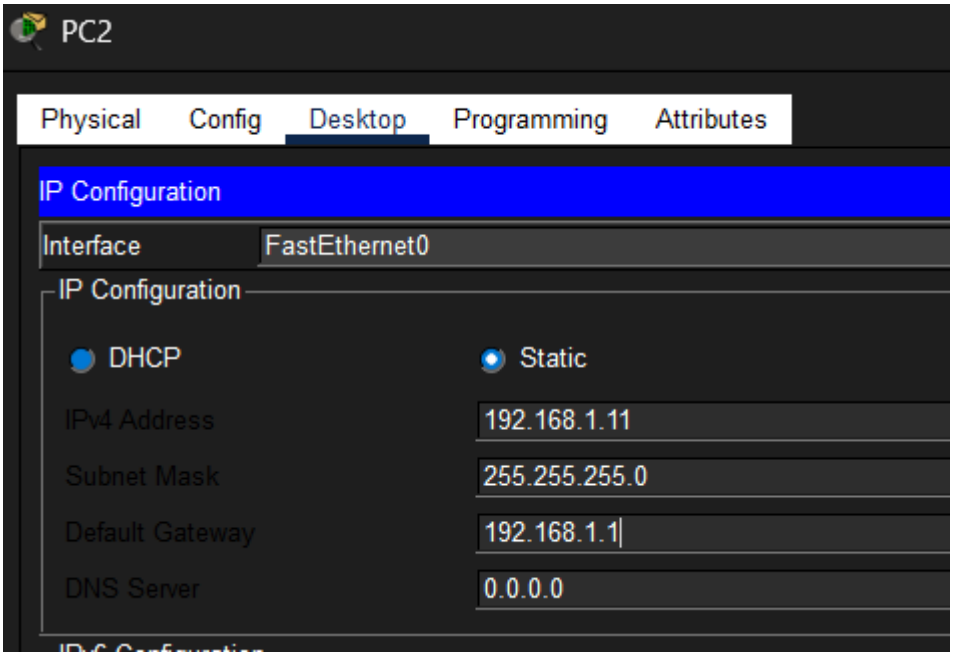
```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

สามารถ ping ไปที่ router ได้สำเร็จเพราะปลายทางอยู่ Subnet เดียวกัน

arp -a

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           000c.cf35.9101        dynamic
```

restore default gateway



PART 5: ADVANCED OBSERVATION

📝 LAB DELIVERABLES

Worksheet 1: Configuration Verification

1)R1 Interface Status

```
R1>show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

2)S1 VLAN Config

```
s1>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

3)PC1 Connectivity

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

4)ARP Resolution

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           000c.cf35.9101        dynamic
  192.168.1.2           0001.96ce.5bcd        dynamic
  192.168.1.11          000a.41e1.c0d7        dynamic
```

Checkpoint	Command Used	Expected Result	Actual Result	Pass/Fail
R1 Interface Status	`show ip int brief`	G0/0 UP/UP	G0/0 UP/UP	Pass
S1 VLAN Config	`show vlan brief`	VLAN 1 Active	VLAN 1 Active	Pass
PC1 Connectivity	`ping 192.168.1.11`	Success (4 replies)	Success	Pass
ARP Resolution	`arp -a` on PC1	2+ entries	3 entries shown	Pass
MAC Learning	`show mac address-table` on S1	2 dynamic entries	3 dynamic entries	Pass

Worksheet 2: Protocol Analysis

Capture this data during simulation:

1. **ARP Request Frame:**

    - Source MAC: 000A-41E1-C0D7

- Destination MAC: FF:FF:FF:FF:FF:FF
- ARP Opcode: 1 request

2. **ICMP Echo Packet:**

   - Time to Live (TTL): 0.004s
   - Protocol Field: ICMP (Value = 1)

3. **Encapsulation Chain:** Data → ICMP Header → IP Header → Ethernet Header → Bits

Worksheet 3: Troubleshooting Journal

Problem Injected	Symptoms Observed	Diagnostic Commands Used	Root Cause	Solution
Wrong Subnet Mask	ไม่สามารถปิงได้	`ping`, `arp -a`	อยู่คนละ subnet mask ทำให้ปิงหากันไม่ได้	แก้ subnet mask ให้ตรงกัน
Interface Shutdown	ไม่สามารถปิงได้	`show ip int brief`	เนื่องจาก Router shutdown	Start router ขึ้นมาใหม่
Incorrect Gateway	ไม่สามารถปิงได้	`ipconfig`, `tracert`	เนื่องจากใส่ default gateway ผิด ทำให้เส้นทาง connect router ไม่ตรง	แก้ default gateway ให้ตรงกับ router จริงๆ

🎓 ASSESSMENT QUESTIONS

1. When PC1 pings PC2, which protocol resolves the MAC address before ICMP operates?
   -ARP (Address Resolution Protocol)

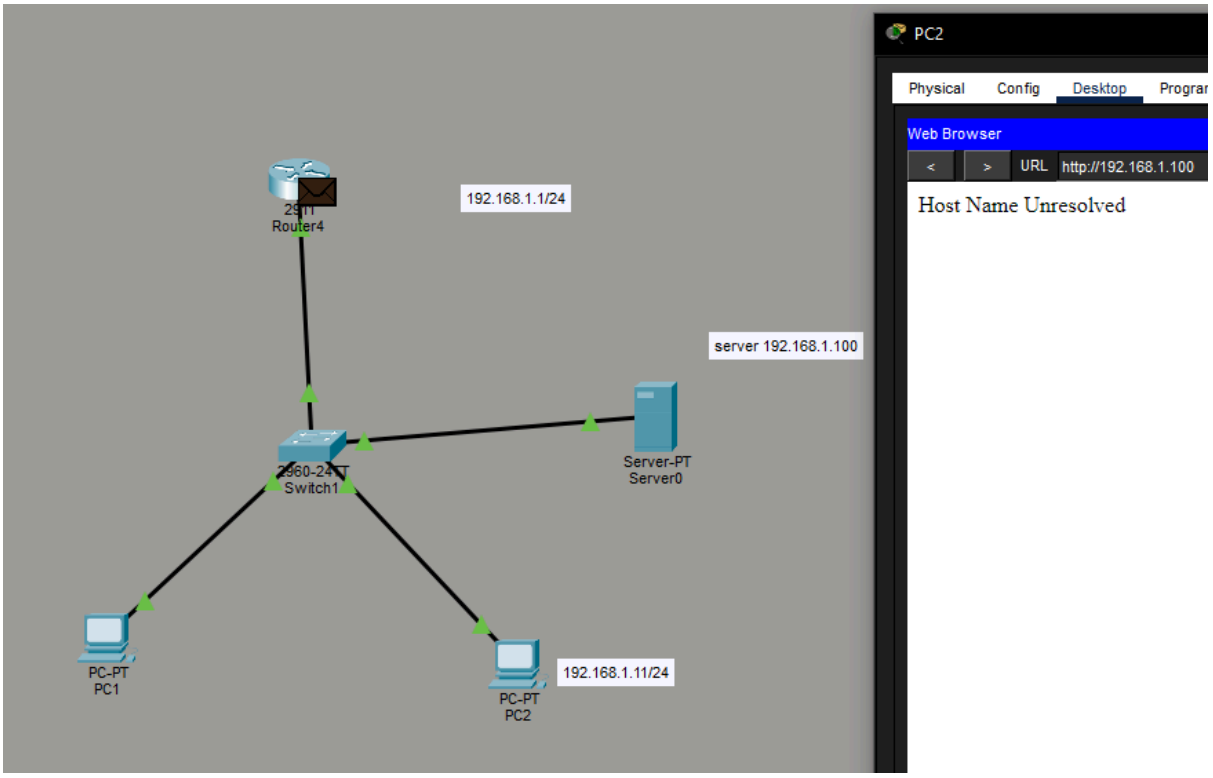
2. What would happen if you changed the subnet mask on PC1 to 255.255.255.128?
   -ไม่มีอะไรเปลี่ยน เนื่องจาก PC1 และ PC2 ยังอยู่ในช่วง subnet เดียวกัน

3. Why does the switch not need an IP address for basic frame forwarding?
   -เพราะสวิตช์ทำงานอยู่ที่ เลเยอร์ 2 และใช้ MAC Address ในการตัดสินใจส่งต่อเฟรม ไม่ได้ใช้ IP Address

4. Which OSI layers are involved when R1 forwards a packet between interfaces?
   -Layer 1 ส่งบิตออกทางสายสัญญาณไปยังอุปกรณ์ถัดไป
   -Layer 2 จัดการเกี่ยวกับโหนดในเครือข่ายเดียวกัน

5. What is the purpose of the TTL field in the IP header you observed?
   -TTL ใน IP Header มีหน้าที่ป้องกันไม่ให้แพ็กเก็ตวนไปเรื่อย ๆ ในเครือข่าย

Bonus



[ITN_Module3_ProtocolLab_File]